USN | | | | | | | | | | [23CSE109]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

COMPUTER NETWORKS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Define the layered operation in networking.	L1	CO 1	**02**
	b.	What are the primary functions of the application layer?	L2	CO 1	**02**
	c.	Name any two features of IPv6 over IPv4.	L1	CO 3	**02**
	d.	Differentiate between TCP and UDP.	L2	CO 2	**02**
	e.	What is DHCP used for in computer networks?	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the operation of the HTTP protocol in the application layer.	L2	CO 1	**05**
		OR			
	b.	Describe the process of electronic mail delivery using SMTP.	L2	CO 1	**05**
3.	**a.**	Illustrate how TCP manages flow control.	L3	CO 2	**05**
		OR			
	b.	Discuss the role of Weighted Fair Queuing (WFQ) in congestion control.	L4	CO 2	**05**
4.	**a.**	Compare static and dynamic IP addressing.	L4	CO 3	**05**
		OR			
	b.	Explain the significance of VLSM in IP addressing.	L2	CO 3	**05**
5.	**a.**	Discuss about VLANs, and how do they improve network management?	L2	CO 4	**05**
		OR			
	b.	Describe the implementation of the Spanning Tree Protocol (STP).	L3	CO 4	**05**
6.	**a.**	Explain the importance of 802.11 security features in wireless networks.	L2	CO6	**05**
		OR			
	b.	Describe the role of DHCP in wireless network operations?	L3	CO 6	**05**

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7.	**a.**	Explain the concept and working of NAT (Network Address Translation) in IPv4 networks.	L2	CO 3	**05**
		OR			
	b.	Discuss the difference between a router and a switch in network layer operations.	L4	CO 3	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss the working of DNS and its importance in the World Wide Web.	L4	CO 1	**10**
		OR			
	b.	Explain the role of FTP in file transfer operations and its benefits.	L2	CO 1	**10**
9.	**a.**	Describe the process of TCP connection establishment with a neat diagram.	L4	CO 2	**10**
		OR			
	b.	Illustrate how Random Early Detection (RED) helps in congestion avoidance.	L5	CO 2	**10**
10.	**a.**	Explain the features of IPv6 and how they overcome the limitations of IPv4.	L4	CO 3	**10**
		OR			
	b.	Describe the working of OSPF in dynamic routing.	L3	CO 3	**10**
11.	**a.**	Discuss the concept of CSMA/CA and how it ensures stability in multiple access networks.	L4	CO 4	**10**
		OR			
	b.	Explain the architecture of a wireless router and its services in a converged WAN.	L5	CO 4	**10**
12.	**a.**	How do antennas play a role in wireless networks? Explain with examples.	L4	CO 6	**10**
		OR			
	b.	Describe the framework and operation of 802.11 networks.	L5	CO 6	**10**
13.	**a.**	Explain the concept and working of NAT (Network Address Translation) in IPv4 networks.	L2	CO 4	**10**
		OR			
	b.	Discuss the difference between a router and a switch in network layer operations.	L4	CO 4	**10**